The electromagnetic spectrum

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Learning target: Relate electromagnetic wavelength, frequency and propagation.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 9.*
50 m; one-way distance is half the round trip.

02. Worked example: explain the model
Shorter wavelength has higher frequency; different EM types do not require different vacuum speeds.

03. Read and interpret the model
Radio, visible, X-ray; X-ray highest frequency.

04. Apply the physics
4 m; halves.

05. Explain your reasoning
EM waves do not require material medium; ordinary sound is pressure motion of matter.

06. Review: connect an earlier idea *Review: Unit 5, Day 12.*
+40 J; internal energy increases.

07. Exit check
No; frequency and wavelength change.

08. STRETCH: test the model
Examples 10^6 Hz and 300 m; 10^9 Hz and 0.3 m.

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01. Warm-up: bring back an earlier idea *Review: Unit 6, Day 9.*
λ=1.2 m; f≈283.3 Hz.

02. Worked example: explain the model
Shorter wavelength has higher frequency; different EM types do not require different vacuum speeds.

03. Read and interpret the model
Radio, visible, X-ray; X-ray highest frequency.

04. Apply the physics
5×10^{14} Hz; 3 m.

05. Explain your reasoning
EM waves do not require material medium; ordinary sound is pressure motion of matter.

06. Review: connect an earlier idea *Review: Unit 5, Day 12.*
Carnot limit η≈0.3333; W_{max}≈266.7 J. Irreversibility and losses reduce actual useful work.

07. Exit check
0.125 m; microwave.

08. STRETCH: test the model
5×10^{14} Hz; 300 nm gives 10^{15} Hz, twice the frequency.